14 The equation of a curve is $y = \frac{16}{x^2} + \frac{3}{x}$.

(a) Determine the coordinates of the point where the curve cuts the x-axis. **[2]**

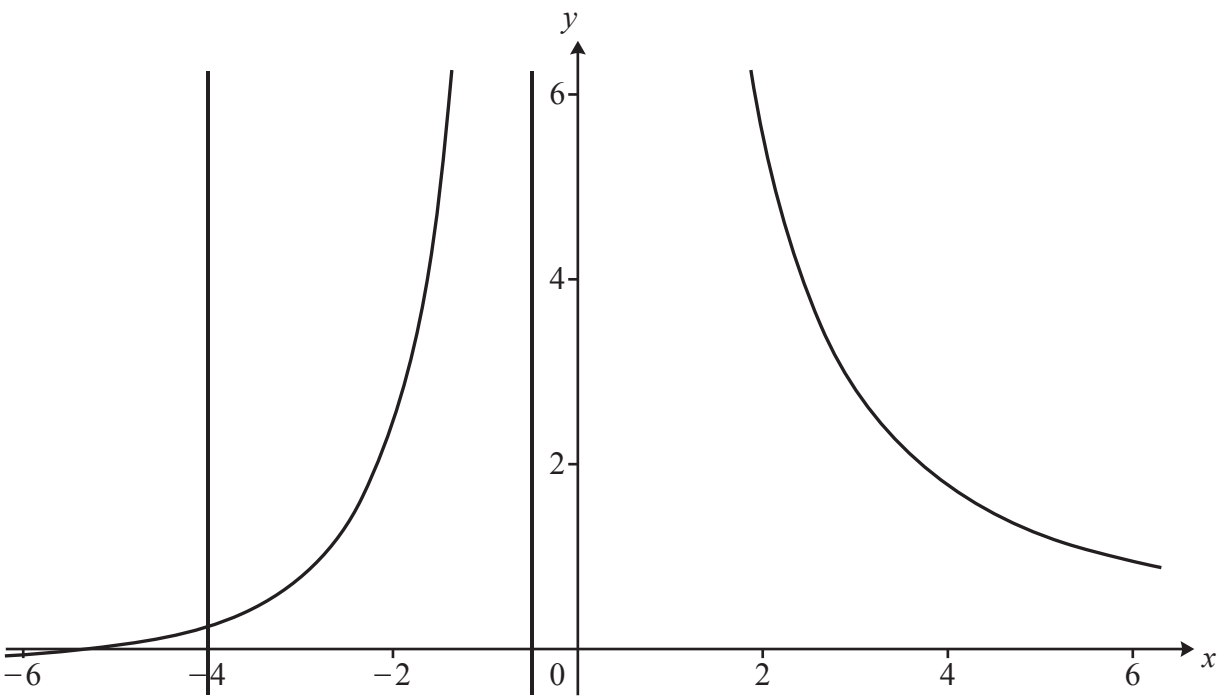
(b) (i) Find $\frac{dy}{dx}$. **[2]**

(ii) Hence determine the exact coordinates of the stationary point on the curve $y = \frac{16}{x^2} + \frac{3}{x}$. **[2]**

(c) (i) Find $\frac{d^2y}{dx^2}$. **[2]**

(ii) Hence determine the set of values of x for which the curve $y = \frac{16}{x^2} + \frac{3}{x}$ is concave downwards. **[2]**

The diagram shows parts of the curve $y = \frac{16}{x^2} + \frac{3}{x}$, the line $x = -4$ and the line $x = -\frac{1}{2}$.



(d) Determine the exact area bounded by the curve $y = \frac{16}{x^2} + \frac{3}{x}$, the x-axis and the lines $x = -4$ and $x = -\frac{1}{2}$. Give your answer in the form $a + b\ln 2$, where a and b are constants to be determined. **[5]**